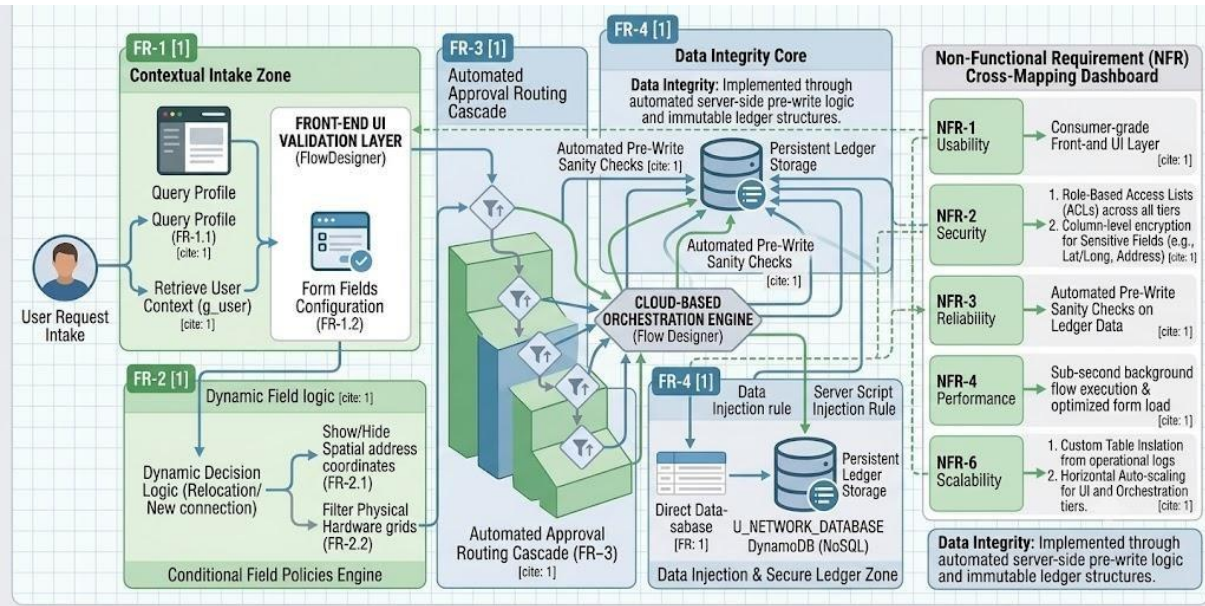


Project Design Phase-II

Solution Requirements (Functional & Non-functional)

Date:	26 July 2026
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Project Name:	Automated Network Request Management System
Maximum Marks:	4 Marks



Functional Requirements:

Following are the functional requirements of the proposed solution.

ID	Functional Area	Detailed Requirement	System Implementation Notes
FR-1.1	Contextual Intake Zone	Query profile and retrieve user context (g_user).	Front-End UI Validation Layer (FlowDesigner).
FR-1.2	Contextual Intake Zone	Dynamically configure form fields.	Managed by the UI validation layer.

ID	Functional Area	Detailed Requirement	System Implementation Notes
FR-2.1	Conditional Field Policies Engine	Dynamically show/hide spatial address coordinates.	Triggered by dynamic decision logic (e.g., Relocation/New Connection).
FR-2.2	Conditional Field Policies Engine	Filter physical hardware grids based on user context.	Linked to dynamic UI and spatial logic.
FR-3	Automated Approval Routing Cascade	Multi-stage automated approval routing paths.	Orchestrated via Cloud-Based Orchestration Engine (Flow Designer).
FR-4.1	Data Injection & Secure Ledger Zone	Perform direct database table write verification.	Feeds directly into U_NETWORK_DATABASE (DynamoDB NoSQL).
FR-4.2	Data Integrity Core	Execute automated server-side pre-write logic and sanity checks.	Ensures immutable ledger structures and data validation before persistent storage.

Non-functional Requirements:

Following are the non-functional requirements of the proposed solution.

NFR ID	Category	Architectural Mapping / Implementation Detail
NFR-1	Usability	Delivery of a consumer-grade Front-end UI Layer.
NFR-2	Security	1. Strict Role-Based Access Lists (ACLs) enforced across all infrastructure tiers. 2. Column-level encryption applied to sensitive fields (e.g., Lat/Long, Address).

NFR ID	Category	Architectural Mapping / Implementation Detail
NFR-3	Reliability	Execution of automated, pre-write sanity checks on all ledger data.
NFR-4	Performance	Sub-second background flow execution and heavily optimized form load times.
NFR-6	Scalability	1. Custom Table isolation to prevent operational logs from degrading performance. 2. Horizontal Auto-scaling capabilities applied to both UI and Orchestration tiers.